Final Project – Stage 2

APMA 2501 - Spring 2026

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Topic for today:

- Converting animated figures into videos

Final Project

Your objective for the final project is to develop a tool that will allow you to **simulate your own three-dimensional, gravity-driven solar system**. Although there is only one simple equation governing all these interactions, successfully coding this simulation will require some careful thinking and consideration of everything we've done this semester.

From Last Time…

Putting it all together

```
1. Set up the structured variable array

2. Calculate the total force on each object

3. Calculate the acceleration; update velocity and position

4. Save the results to a movie file (today!!!)
```

FAQs So Far

Q: My planets always move apart, what's the problem?

A: Force is a vector, and the sign is important to us. When you calculate the separation in the x, y, and z direction, make sure you've got the +/- correct.

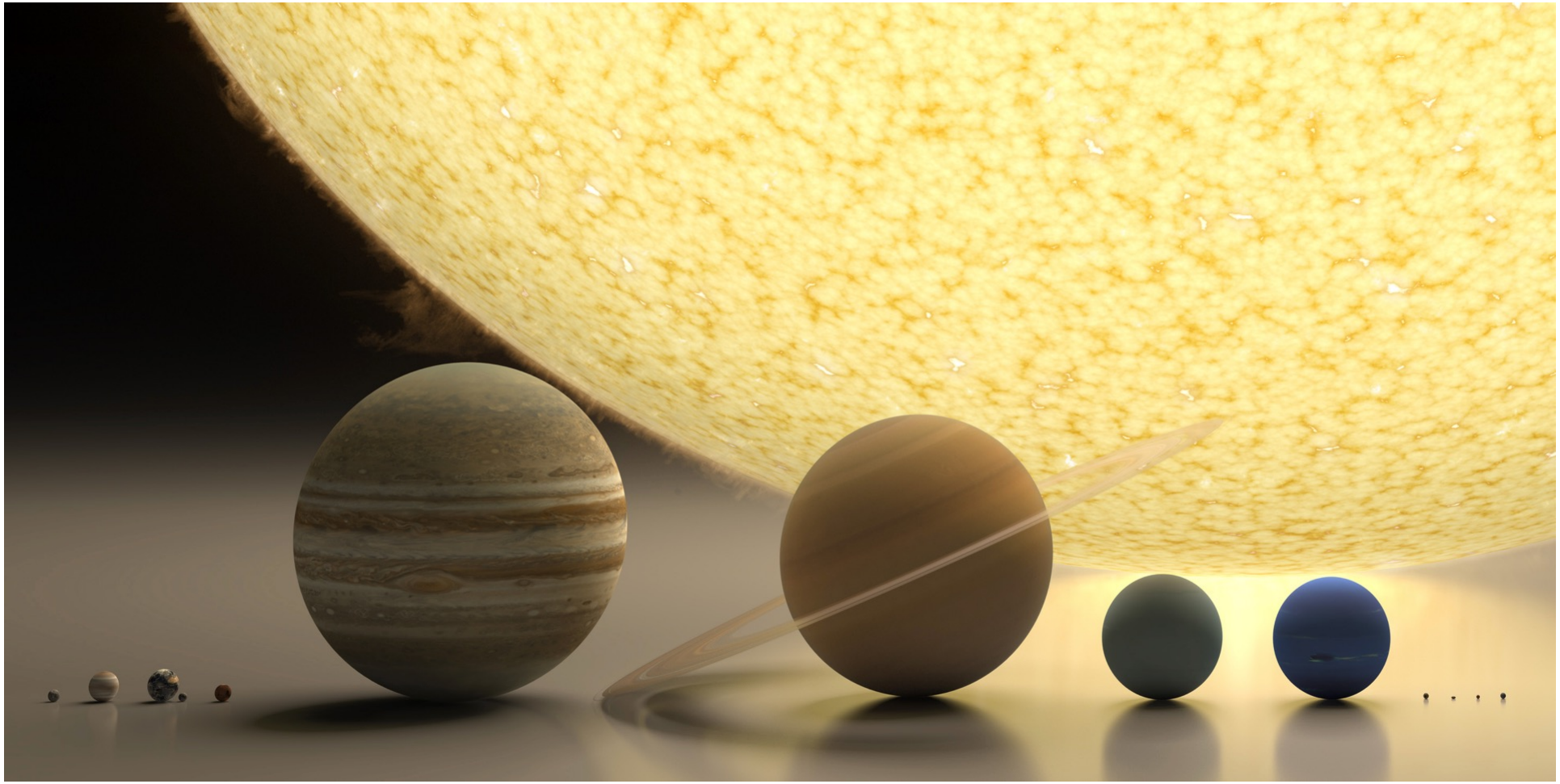
Q: How does hold on work?

A: `hold on` retains the current plot so that subsequent plotting commands add to the existing figure (instead of overwriting it) allowing us to accumulate features one at a time. There are good examples in the documentation, `doc hold`.

Q: I am not seeing the wobble of the sun in my simulator, did I do something wrong?

A: No, because the mass of the sun is so large, which seems very stable in the simulation.

A Note on Scaling



Saving Movie Files

The **drawnow** method of animating is nice, but doesn't create a sharable, permanent movie. Let's consider creating a movie file using a very simple example: a traveling sine wave.

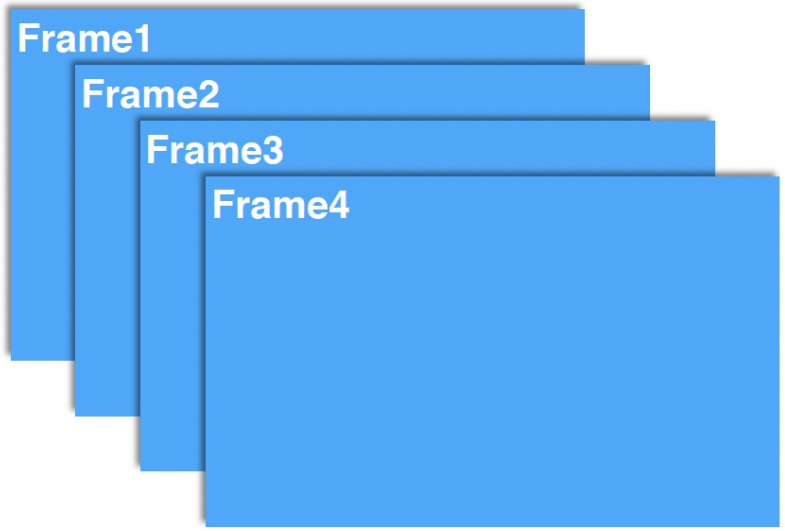
```
% STEP 1: Create the video object for writing
vid = VideoWriter('SineWave', 'MPEG-4');

% STEP 2: Set some movie options (help VideoWriter for more)
vid.FrameRate = 30;
vid.Quality = 100;

% STEP 3: Open the video file for writing
open(vid);
```

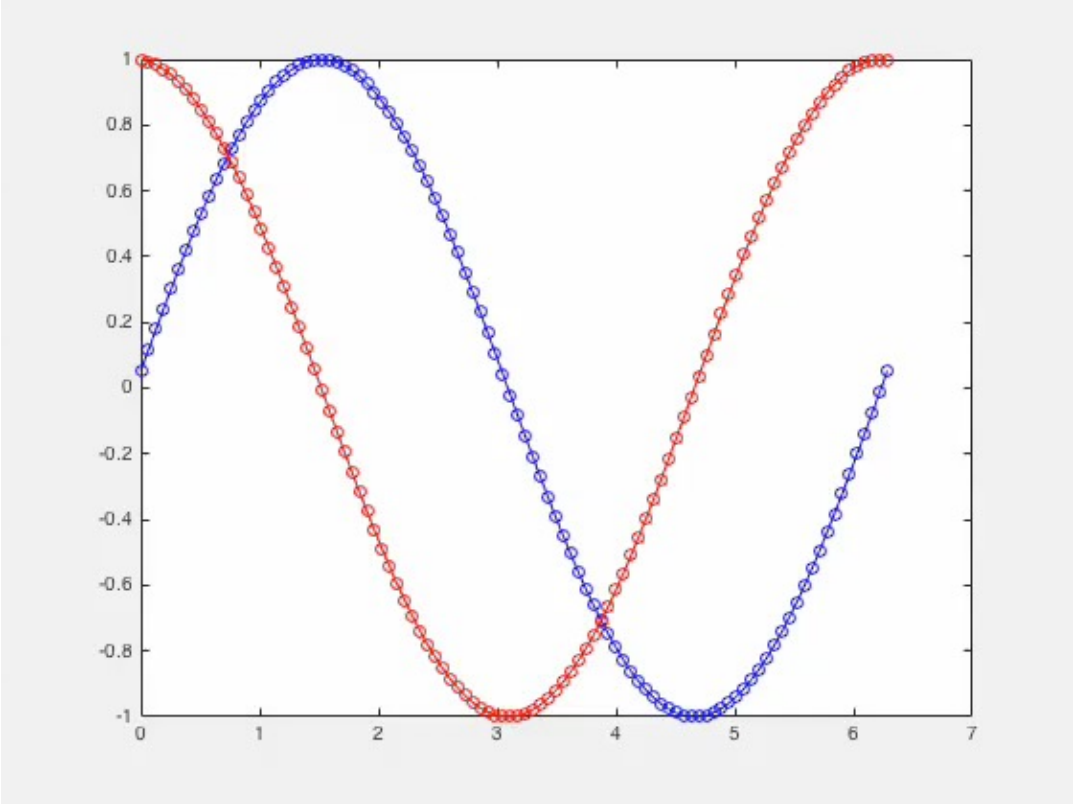
File Name File Type

Saving Movie Files




```
% STEP 4: Add current plot as an
% additional frame in the movie
writeVideo(vid, getframe);

% STEP 5: Close the video file when
% finished
close(vid);
```



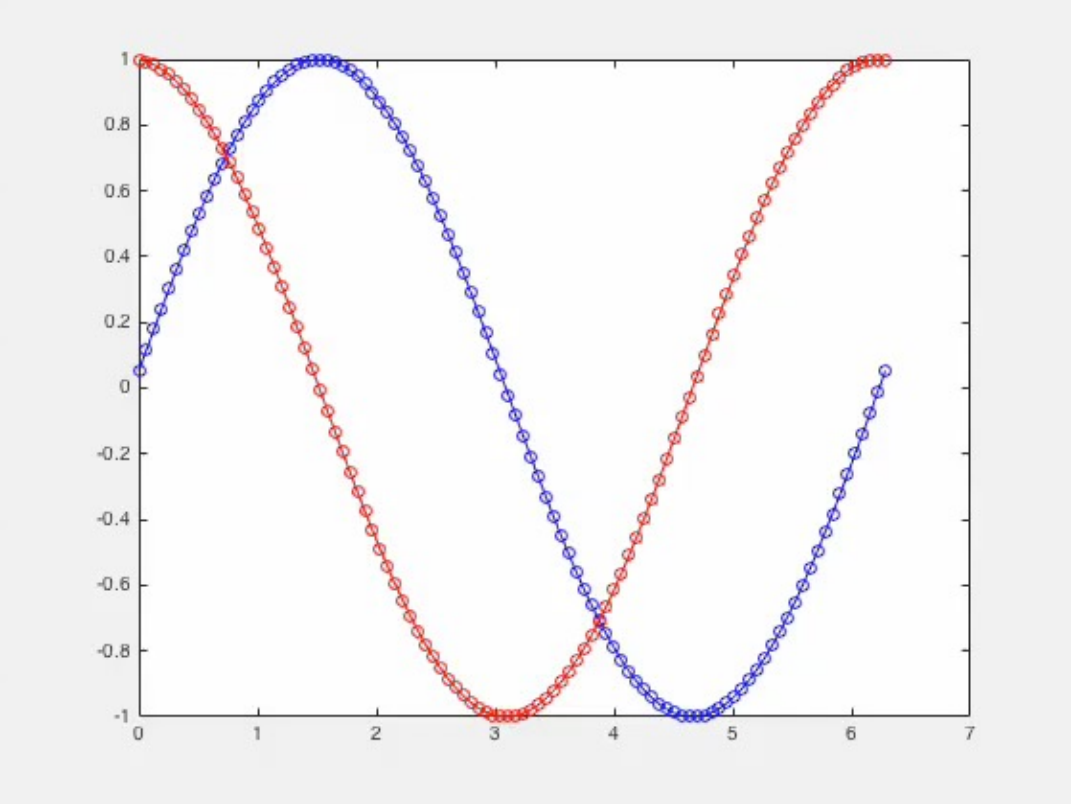
Saving Movie Files: Cheat Sheet

For Testing (doesn't create movie file)

- `drawnow`

Basic Video-Writing Functions

- `VideoWriter`
- `open/close file`
- `writeVideo`
- `getframe`



Now we have learned how to make videos

Putting it all together

```
1. Set up the structured variable array

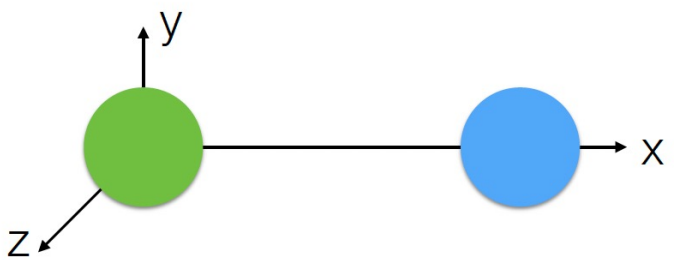
2. Calculate the total force on each object

3. Calculate the acceleration; update velocity and position

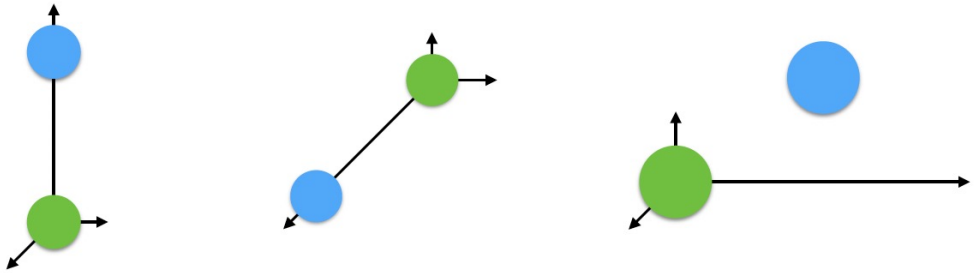
4. Save the results to a movie file (done!)
```

Final Project Tips

1. Start Small — much easier to debug!



2. Build up **slowly**.



3. Print statements and hand calcs.

Thank You!

